

HIGHLIGHTS OF QUALIFICATIONS

- Deep expertise across the model life cycle — development, validation and effective challenge — in credit portfolio risk, Economic Capital and stress testing.
- Proficient in programming Python (NumPy, Pandas, Polars, Jupyter, FastAPI), SQL, R (tidyverse, Shiny), SAS and Matlab. Developed various portfolio risk models & analytical tools, accessible at <https://jason2003wxy.github.io>, demonstrating expertise in statistical model development, programming and visualisation.
- Hands-on with agentic AI: designed and operates a Claude Code-driven investment-research knowledge base (240+ company reports, industry KPI templates, instruction-level governance) — firsthand experience with agentic workflow design and observed failure modes (hallucination, citation drift, instruction-following gaps).

Work Experience

2018 – Present **Stress Testing & Capital Analytics**, VP, SRM, Credit Suisse International

- Responsibilities*
- As entity SME, bridge the gaps between UK regulatory requirements and Group Economic Capital & stress testing modelling capabilities. Provide analytical and modelling expertise to entity senior management, facilitating sign-off of stress scenario loss, RWA and Pillar 2A capital add-ons for UK ICAAP.
 - Supervise offshore analytical team, successfully led their transition from Excel to R and Python.
- Achievement*
- Corrected and replaced Loan & Advance and CCR portfolio credit stress loss and RWA, and Standardised CVA RWA calculations, which were previously performed manually on Excel by multiple Risk and Finance teams. The new stress testing platform is implemented in Python. Furthermore, the trade level risk sensitivities (e.g., DV01, CS01 in order of millions) are sourced from the back-end Credit Analytics database (SQL executed directly in Python) and then restructured, cleaned and aggregated at the specified counterparty-product level for explanatory analyses.
 - The developed Python solution also includes automated methodology documentation with explanatory analyses (Jupyter Notebook) such as period-on-period analysis, SQLite DB for historical data storage, and version control (Atlassian Bitbucket) and issue tracking (JIRA), which has increased efficiency and accuracy and transparency (audit trails) drastically. Mentoring offshore junior team member for code maintenance and execution.
 - Developed & implemented (Python) stress SA-CCR EAD methodology for ICAAP stress testing in response to PRA SREP feedback.
 - Developed & implemented (Python) stress rating methodology based on BASEL Merton model to translate expert opinions (economists and credit industry / regional analysts) on rating behaviour under stress narratives into stressed migration matrices. Coached offshore junior team member to host the working group periodically.
 - Developed & implemented (Python) a stress testing methodology quantifying Additional Termination Event (ATE) based on stressed migration matrices in response to PRA SREP feedback.
 - Developed & implemented (Python) scenario expansion methodology based on empirical distribution to convert economic/financial shocks into various frequencies (e.g., annually, monthly). The methodology is also used for shock severity assessment by retrieving historical outcomes according to the shock empirical distribution.
 - For capital management stakeholders, developed interactive (R Shiny) web-app to visualise capital adequacy implied by the stress testing, interactively perform what-if analysis for capital & financial plan assumption sensitivity. Implemented drill-down and PoP comparison features to identify the capital plan components driving the Pillar 2 capital buffer.
 - Developed management dashboard (R Shiny) for senior risk management to visualise Capital, RWA, Leverage Ratio and Liquidity metrics with *drill-down* and *PoP* comparison capabilities. Coached offshore team members to compile monthly capital risk report for entity Risk Committee using the dashboard.

2015 – 2017 **EC & Stress Testing**, *VP/AD*, ERM, Royal Bank of Canada (Europe Ltd)

- Achievements*
- Led the development of Stress Testing and Economic Capital methodologies for UK and multiple EU entity ICAAP and BAU stress testing practices.
 - Developed & implemented methodologies to quantify credit concentration risk (single name, sector & region, loan product) for a number of European legal entities. This result has been used for estimating and reporting their Pillar 2A capital add-on. Implementation was done in Python (Jupyter) with input from vendor model (Moody's Risk Frontier).
 - Developed & implemented methodologies to quantify Pillar 2A operational risk capital add-on for a number of European legal entities.
 - Developed & implemented statistical models to project business unit balance sheet, net interest income and non-interest income under different stress scenarios.
 - Developed & implemented management dashboard using R Shiny, allowing risk managers to visualise ultra-high-net-worth (UHNW) real estate portfolio on the map of London and other prime locations. The dashboard incorporated the Savills indices for revaluation and stress credit parameters sourced from the Risk system. Furthermore, it provides what-if scenario analysis to capture impact of the Stamp Duty Land Tax on mortgage LTV and obligor debt service ability.

2011 – 2015 **Economic Capital Modelling**, *Model Developer*, Royal Bank of Scotland Group

- Achievements*
- Developed & implemented methodology to quantify credit concentration risk. The approach has been accepted by the UK regulator and successfully reduced 2 billion GBP regulatory capital requirement for RBS Group.
 - As a founding team member, developed in-house Economic Capital models from scratch for RBS Group; outstanding performance led to progressive promotions.
 - Developed multi-factor Correlation model of Credit EC simulation engine, the factor model estimates correlations among different sectors/industries and regions based on equity and CDS time series (implemented in MATLAB).
 - Developed economic capital methodology for defaulted assets, which captures concentration risk and correlation among different asset classes (implemented in MATLAB).
 - Implemented EC tools to quantify portfolio name and industrial/sector concentration risk. These tools have been used across multiple subsidiaries for quantifying credit concentration capital add-ons required by local regulators. Implementation was done in SAS, MATLAB and C++.
 - Developed integrated top-down stress testing framework. The time series models translate economic factor impact to bank's net interest incomes, non-interest incomes, and operational risk losses. The implementation was done in R and L^AT_EX for automated documentation.

2009 – 2011 **Model Validation**, *Quantitative Risk Analyst*, ABN AMRO Group

- Performed independent model validation of regulatory credit risk models (PD, LGD, EAD) and credit scoring models (mortgage / credit card application).
- Presented findings to the senior risk management committee.

Education

2011 **Econometrics**, *University of Groningen*, Netherlands
BSc (honors distinction)

2017 **Machine Learning**, *Stanford University via Coursera*
Certificate: Machine Learning by Andrew Ng

Interests

Triathlon IM70.3 (Valencia 5:45); Cycling (RideLondon 5:05); Running (Hackney Half 1:50)

Coding Please visit <https://jason2003wxy.github.io> for my work gallery

AI / agentic workflows Personal knowledge-base project using **Claude Code** (agentic AI coding assistant) and **Obsidian**: ingested 240+ company annual/quarterly reports into a structured wiki with industry-specific KPI templates (banks, oil & gas, manufacturing) and instruction-level governance rules. Hands-on with agentic workflow design and the practical failure modes of LLM / agent systems (hallucination, citation drift, instruction-following).